BKZ 2.0: Better Lattice Security Estimates

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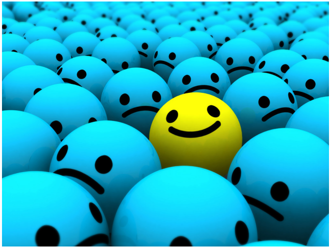
Asiacrypt, 2011

Introduction

Advantages of *Lattice-Based Cryptography*

- Security based on worst-case hardness
- Potential resistance to quantum sub-exponential attacks
- Asymptotic efficiency
- New functionalities (Fully-Homomorphic Encryption, etc)



Current Situation

Current Situation: Very few *concrete proposals* for lattice-based cryptosystems, partly due to the difficulty of lattice security estimates.

To give better security estimates, we need to predict the behavior of the best lattice reduction algorithms.

Our contribution

BKZ reduction is the best reduction algorithm known in practice.

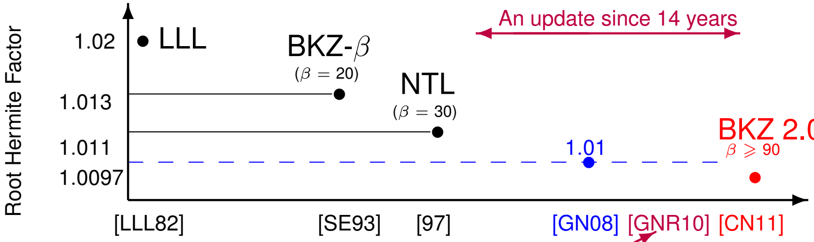
But all lattice security estimates rely on NTL's 1997 implementation of BKZ.

BKZ 2.0 We implemented a state-of-the-art BKZ, including recent algorithmic improvements such as [GNR10] pruning.

Better Lattice Security Estimates We present a simulation algorithm, which allows to predict the performances of high-blocksize BKZ, like BKZ-140.

Chronology

We provide a State-Of-The-Art BKZ implementation..



..incorporating Gama-Nguyen-Regev pruning.

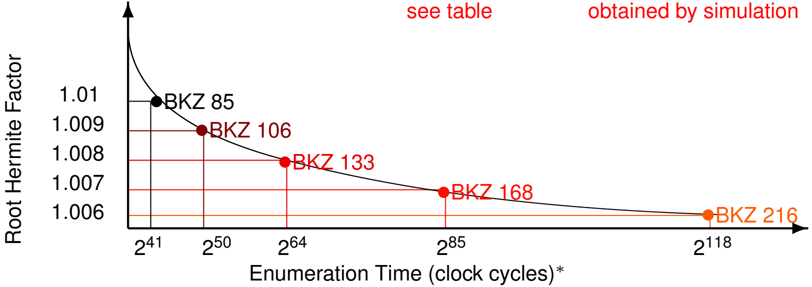
Contributions For The Future

We make it possible to do more precise security estimates.

Target System : Lattice basis , Critical quality .

Security Level ≈ Dim × Enum Time × Number of Rounds .

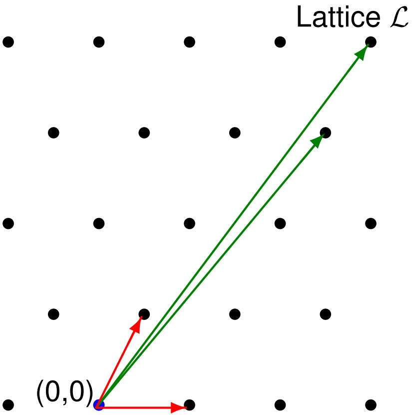
see table obtained by simulation



Outline

Lattice Basis

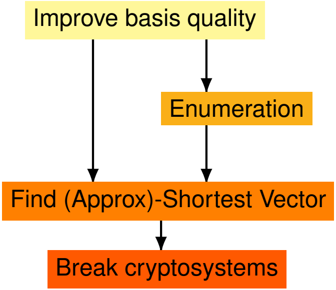
Basis with good/bad quality.



Reduction

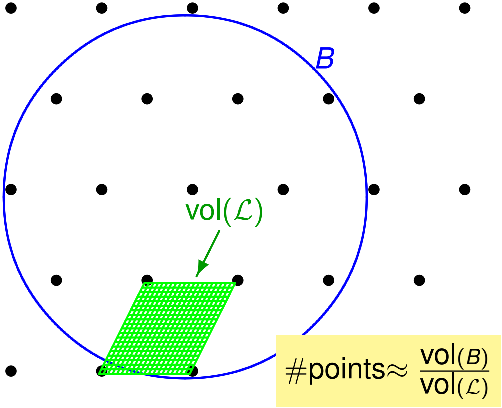
$$L_2 \xrightarrow{\quad} L_1$$

$$\begin{pmatrix} 377 & 610 \\ 178 & 288 \end{pmatrix} \quad \begin{pmatrix} 2 & 0 \\ 3 & 2 \end{pmatrix}$$



Gaussian Heuristic

$\mathcal{L}$: A random lattice of dimension n



$\text{Vol}(\mathcal{L}) = \det(L)$

Root Hermite Factor
$= (\|b_1\|/\text{vol}(\mathcal{L})^{1/n})^{1/n}$

Gaussian Heuristic predicts length of shortest vector $\|v\|$:
$|v| \approx \text{radius}(B)$ s.t. $\text{vol}(B) = \text{vol}(\mathcal{L})$ for dimension > 50.

LLL Reduction in dimension n

$$\begin{pmatrix}
\|b_1^*\| & 0 & 0 & \dots & & 0 & 0 \\
\star & \|b_2^*\| & 0 & \dots & & 0 & 0 \\
\star & \star & \ddots & 0 & & \vdots & \vdots \\
\star & \star & \|b_i^*\| & 0 & & & \\
\star & \star & \star & \|b_{i+1}^*\| & & & \\
\star & \star & \star & \star & \star & \ddots & 0 \\
\star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

LLL reduces using 2-dim blocks.

LLL Reduction in dimension n

$$\begin{pmatrix} \|b_1^*\| & 0 & 0 & \dots & & 0 & 0 \\ \star & \|b_2^*\| & 0 & \dots & & 0 & 0 \\ \star & \star & \ddots & 0 & & \vdots & \vdots \\ \star & \star & \|b_i^*\| & 0 & & & \\ \star & \star & \star & \|b_{i+1}^*\| & & & \\ \star & \star & \star & \star & \star & \ddots & 0 \\ \star & \star & \star & \star & \star & \star & \|b_n^*\| \end{pmatrix}$$

LLL reduces using 2-dim blocks.

BKZ Reduction in dimension n

$$\begin{pmatrix}
\|b_1^*\| & 0 & \ldots & & & & & 0 & 0 \\
\star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\
\star & \star & \|b_3^*\| & 0 & & & & & \\
\star & \star & \star & \ddots & 0 & & & & \\
\star & \star & \star & \star & \|b_\beta^*\| & & & & \\
\star & \star & \star & \star & \star & \ddots & & & \\
\star & \star & \star & \star & \star & \star & & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

The 1st block

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$
\begin{array}{cc}
\begin{pmatrix}
\|b_1^*\| & 0 & \cdots & & & & & 0 & 0 \\
\star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\
\star & \star & \|b_3^*\| & 0 & & & & & \\
\star & \star & \star & \ddots & 0 & & & & \\
\star & \star & \star & \star & & & & & \\
\star & \star & \star & \star & \star & \|b_{\beta+1}^*\| & & & \\
\star & \star & \star & \star & \star & \star & \ddots & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}
&
\begin{array}{l}
\text{The} \\
\text{2nd} \\
\text{block}
\end{array}
\end{array}
$$

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$\begin{pmatrix} \|b_1^*\| & 0 & \dots & & & & & 0 & 0 \\ \star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\ \color{red}\star & \color{red}\star & \color{red}\|b_3^*\| & 0 & & & & & \\ \star & \star & \star & \ddots & 0 & & & & \\ \star & \star & \star & \star & & & & & \\ \star & \star & \star & \star & \star & \ddots & & & \\ \star & \star & \star & \star & \star & \star & \|b_{\beta+2}^*\| & & \\ \star & \star & \star & \star & \star & \star & \star & \ddots & \\ \star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\| \end{pmatrix}$$

The 3rd block

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$
\begin{pmatrix}
\|b_1^*\| & 0 & \dots & & & & & 0 & 0 \\
\star & \ddots & 0 & & & & & \vdots & \vdots \\
\star & \star & \ddots & 0 & & & & & \\
\star & \star & \star & \ddots & & & & & \\
{\color{red}\star} & {\color{red}\star} & {\color{red}\star} & {\color{red}\star} & {\color{red}\|b_{n-\beta+1}^*\|} & 0 & & & \\
\star & \star & \star & \star & \star & \ddots & & & \\
\star & \star & \star & \star & \star & \star & & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}
$$

The tail

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$\begin{pmatrix}
\|b_1^*\| & 0 & \dots & & & & & 0 & 0 \\
\star & \ddots & 0 & & & & & \vdots & \vdots \\
\star & \star & \ddots & 0 & & & & & \\
\star & \star & \star & \ddots & & & & & \\
\star & \star & \star & \star & \ddots & & & & \\
\color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}\|b_{n-\beta+2}^*\| & 0 & & \\
\star & \star & \star & \star & \star & \star & & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

Smaller blocks for the tail

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$\begin{pmatrix}
\|b_1^*\| & 0 & \dots & & & & & 0 & 0 \\
\star & \ddots & 0 & & & & & \vdots & \vdots \\
\star & \star & \ddots & 0 & & & & & \\
\star & \star & \star & \ddots & & & & & \\
\star & \star & \star & \star & \ddots & & & & \\
\star & \star & \star & \star & \star & & & & \\
\star & \star & \star & \star & \star & \star & \ddots & & \\
\color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}\star & \color{red}{\|b_{n-1}^*\|} & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

End of the round

BKZ reduces using β-dim blocks by enumeration.

BKZ Reduction in dimension n

$$\begin{pmatrix}
\|b_1^*\| & 0 & \cdots & & & & & 0 & 0 \\
\star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\
\star & \star & \|b_3^*\| & 0 & & & & & \\
\star & \star & \star & \ddots & 0 & & & & \\
\star & \star & \star & \star & \|b_\beta^*\| & & & & \\
\star & \star & \star & \star & \star & \ddots & & & \\
\star & \star & \star & \star & \star & \star & & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

The 2nd round

One round uses $\approx n$ enumeration.

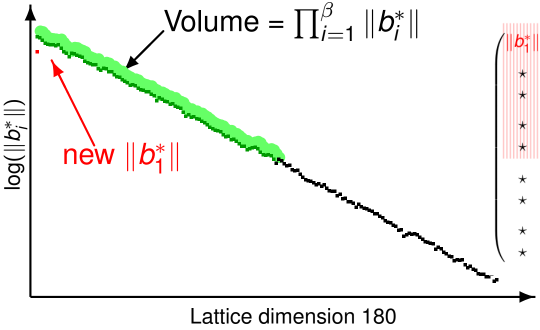
Arising question

In practice,

- How does the basis evolve as the number of rounds increase?

Simulating Quality Of Basis

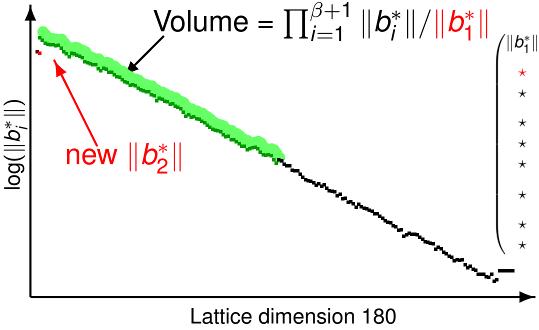
$\|b_i^*\|$ can be predicted in practice by Gaussian heuristic, except for a few blocks.
Local blocks tend to behave like random lattices, as the blocksize increases.



$$\begin{pmatrix}
\|b_1^*\| & 0 & \cdots & & & & & 0 & 0 \\
\star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\
\star & \star & \|b_3^*\| & 0 & & & & & \\
\star & \star & \star & \ddots & 0 & & & & \\
\star & \star & \star & \star & \|b_\beta^*\| & & & & \\
\star & \star & \star & \star & \star & \ddots & & & \\
\star & \star & \star & \star & \star & \star & & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

Simulating Quality Of Basis

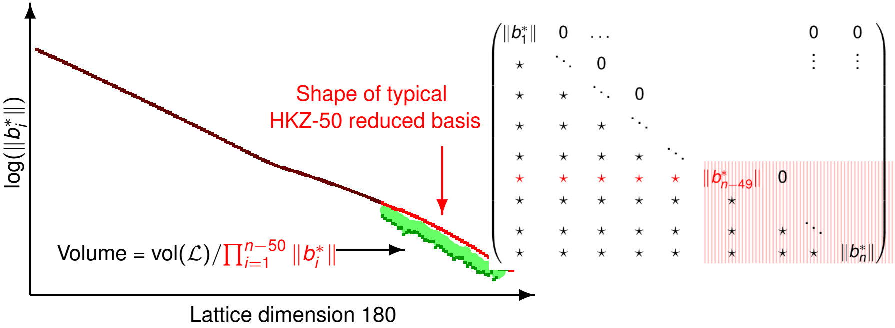
To apply Gaussian Heuristic,
we just need to know the volume.



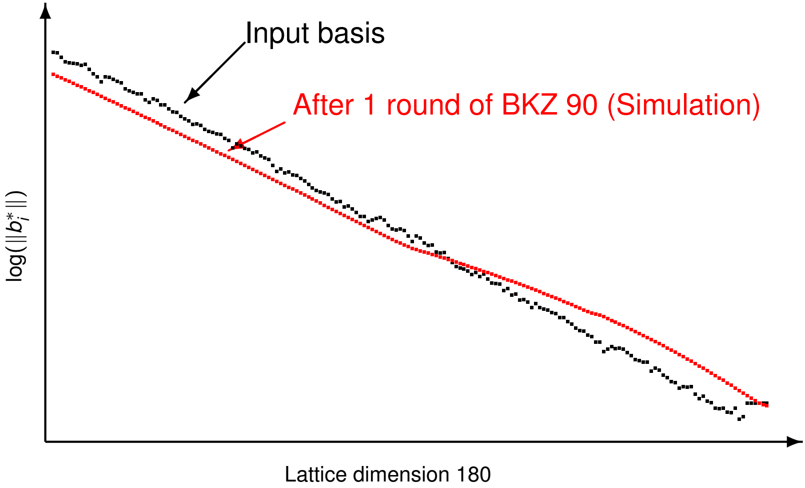
$$\begin{pmatrix}
\|b_1^*\| & 0 & \cdots & & & & & 0 & 0 \\
\star & \|b_2^*\| & 0 & & & & & \vdots & \vdots \\
\star & \star & \|b_3^*\| & 0 & & & & & \\
\star & \star & \star & \ddots & 0 & & & & \\
\star & \star & \star & \star & & & & & \\
\star & \star & \star & \star & \star & \|b_{\beta+1}^*\| & & & \\
\star & \star & \star & \star & \star & \star & \ddots & & \\
\star & \star & \star & \star & \star & \star & \star & \ddots & \\
\star & \star & \star & \star & \star & \star & \star & \star & \|b_n^*\|
\end{pmatrix}$$

Simulating Quality Of Basis

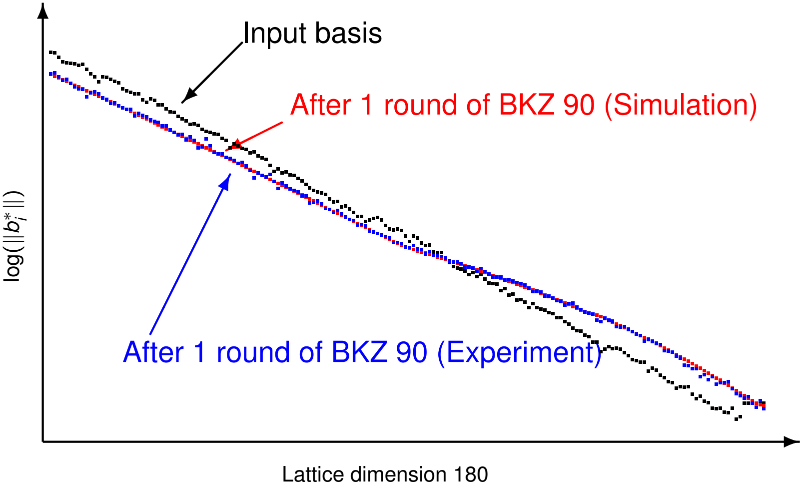
For the "tail" where block size $\leqslant 50$,
we assume that big block is a typical HKZ-50 reduced basis.



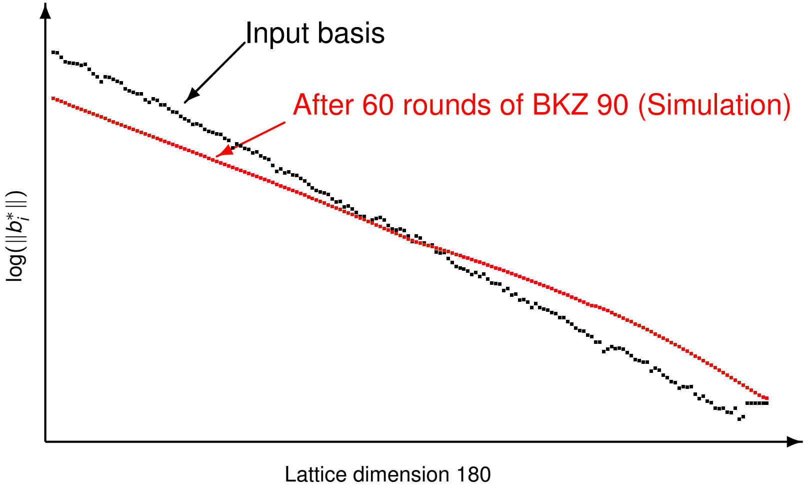
Simulating BKZ Process



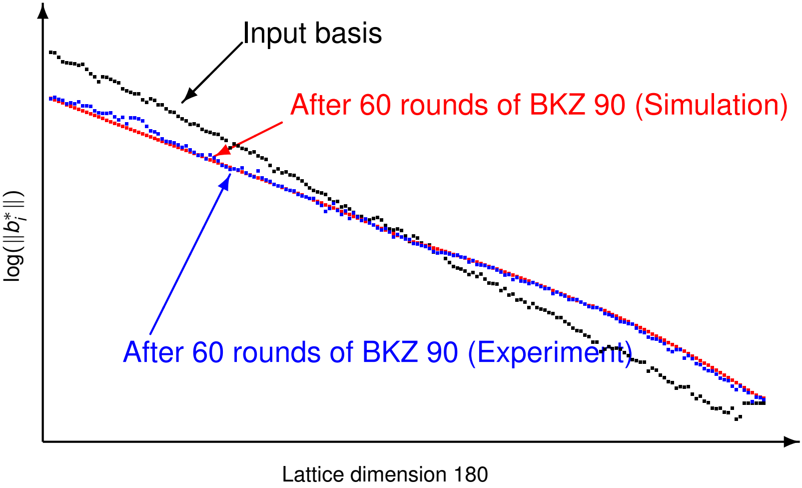
Simulating BKZ Process



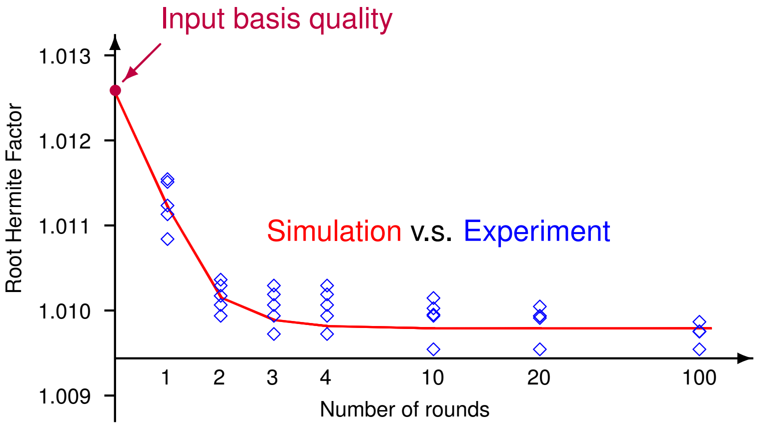
Simulating BKZ Process



Simulating BKZ Process



Simulating BKZ Process

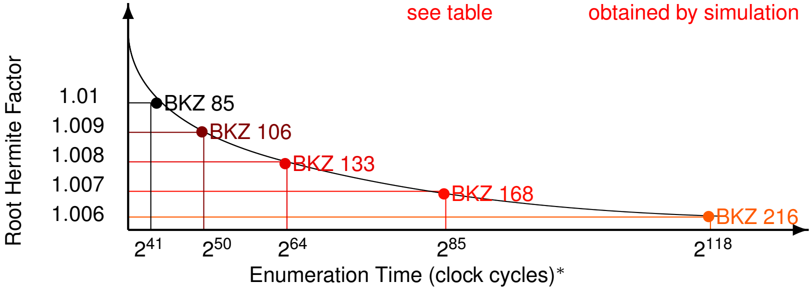


Security Estimates

Target System : Lattice basis , Critical quality .

Security Level ≈ Dim × Enum Time × Number of Rounds .

see table obtained by simulation



Example 1: Application to NTRUsign-157

NTRUsign-157

- Critical root-Hermite-factor: 1.0089.
- Lattice dimension: 314.
- Require BKZ-110 for $\approx$ 6 rounds.
- Enumeration time for dimension 110: 2^{52} clock cycles.

Security Level $\approx$ Dim $\times$ Enum Time $\times$ Number of Rounds

$314 \times 2^{52} \times 6 \approx 2^{63}$ clock cycles.

Previous security estimates by NTRU: 2^{93}.

That's a huge gap.

Example 2: Application to Gentry-Halevi's FHE schema

Gentry-Halevi's FHE schema, large challange

- Critical root-Hermite-factor: 1.0081
- Lattice dimension: 32768.
- Require BKZ-130 for $\approx$ 60000 rounds.
- Enumeration time for dimension 130: 2^{67} clock cycles.

Security Level $\approx$ Dim $\times$ Enum Time $\times$ Number of Rounds

$32768 \times 2^{62} \times 60000 \approx 2^{95}$ clock cycles.

Our Improvements To BKZ

Pruning	Speed up enumeration (GNR Euro10)
Early-Abort	The first few rounds do the majority of work.
Preprocessing	Improve local basis quality before enumeration.
Others	Bound enumeration radius, etc.

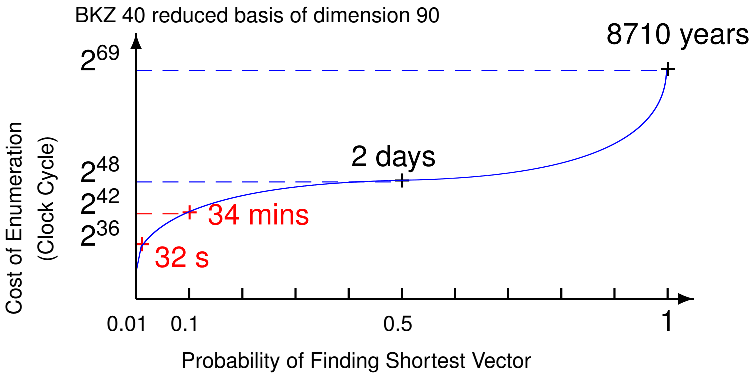
Implementations are tricky.

Significant computational effort: several core-years.

Sound Pruning in BKZ reduction

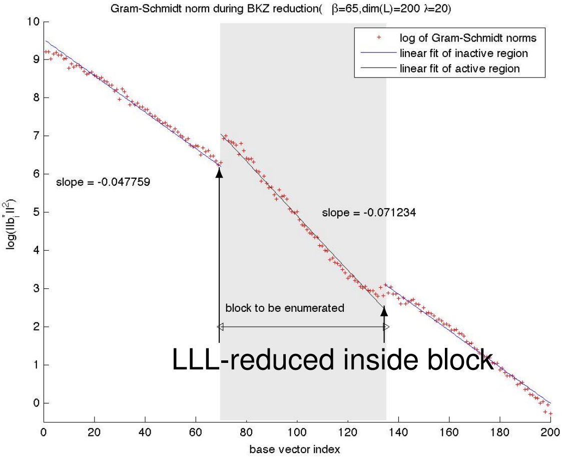
Pruning speeds up enumeration exponentially.

Choose optimized trade-off between success probability and cost.



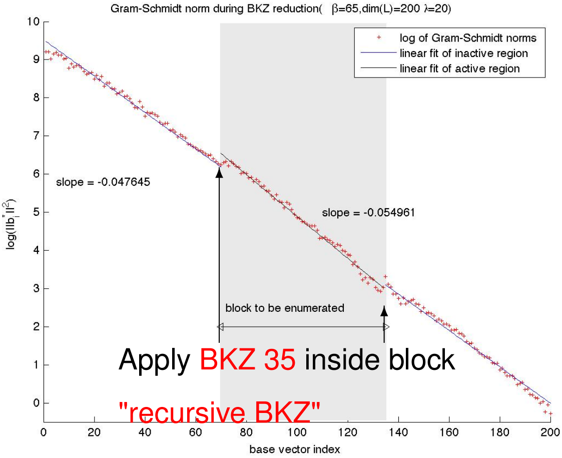
Local Basis Preprocessing

The quality inside a block impacts running time of enumeration.



Local Basis Preprocessing

Improving the local basis speeds up enumeration.



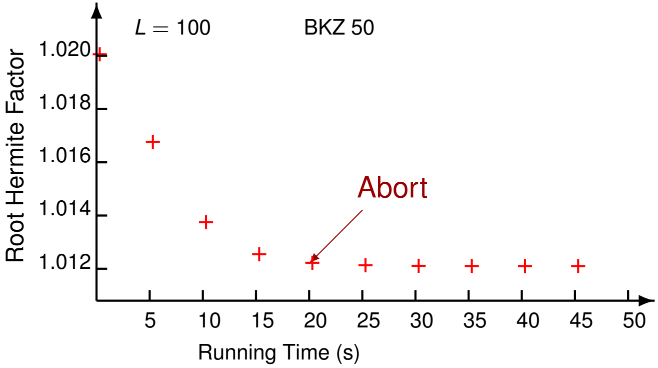
Apply BKZ 35 inside block

"recursive BKZ"

Abort BKZ reduction after a number of rounds

Most of the work is done in early rounds.

We run BKZ for fixed number of round without much loss of quality.



Summary

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Better Lattice Security Estimates We present a simulation algorithm, which allows to predict the performances of high-blocksize BKZ.

For more details please refer to full version of our paper.

Thank you!

감사합니다 謝謝!

谢谢! Cảm ơn bạn! *Danke!*

شكرا لك! ありがとう!

Merci! Σας ευχαριστούμε! ¡*Gracias*!